33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10006

F23-SR-002
415V FINAL TAILS PUMPS SWITCHROOM

F23-TR-023
TRANSFORMER ROOM

F23-TR-023
33/0.433kV
1000kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

1600/ 1A
2.5VA
5P20

F23-SW-023
FINAL TAILS PUMPS 415V MOTOR CONTROL CENTRE

1600/ 5A
5VA
CL 1

1600/ 5A
20VA
CL 0.5

00F-01
00F-08
TEST
PM
20A
63A
2A
63A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
1600A
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
100A
100A
SD
NOTE 6

415V 3NE~ 50Hz 1600A 65kA for 1sec

| 2R-11 | 3R-11 | 2F-01 | 3F-01 | 3F-05 | 3F-09 INC | 1R-01 | 3F-13 |
|---|---|---|---|---|---|---|---|
| 400A | 400A | 20A | 20A | 20A | 20A | 630A | 160A |
| 2A 32A | 2A 32A | 2A 32A | 2A 32A | 2A 32A | 2A 32A | 2A 32A | 2A 32A |
| 250A | 250A | 70A | 70A | 70A | 70A | 400A | 70A |
| EOL | EOL | EOL | EOL | EOL | EOL | 160A 160A | |

CONTINUED ON SHT 2
2000-F23-SLD-EL-10202

FEEDER TO F23-SW-123
2000-F23-SCM-EL-10209

E-SHUT
F23SR002-LCS
NOTE 12

| F00-PS-464 | F00-PS-465 | F23-FN-462 | F23-FN-463 | F23-FN-464 | F23-FN-465 |
|---|---|---|---|---|---|
| EVENT POND SUBMERSIBLE PUMP | SILT TRAP SUBMERSIBLE PUMP | FINAL TAILS PUMP 1 MOTOR COOLING FAN | FINAL TAILS PUMP 2 MOTOR COOLING FAN | FINAL TAILS PUMP 3 MOTOR COOLING FAN | FINAL TAILS PUMP 4 MOTOR COOLING FAN |
| 2000-F00-SCM-EL-10211 | 2000-F00-SCM-EL-10212 | 2000-F23-SCM-EL-10218 | 2000-F23-SCM-EL-10219 | 2000-F23-SCM-EL-10220 | 2000-F23-SCM-EL-10221 |
| 132kW | 132kW | 3.7kW | 3.7kW | 3.7kW | 3.7kW |

160A
F23-DB-101
FINAL TAILS PUMPS
11kV SWITCHROOM
HV VSD AUXILIARY SUUPLIES DB
2000-F23-SCM-EL-10224

| Approved | X |
|---|---|
| Approved as Noted | |
| Not Approved | |
| For Information Only | |

Signed [redacted] Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

## NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F23-SW-023 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F23-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

## LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

# 2000-F23-SLD-EL-10201_Rev1

Final tailings handling & storage F23-SW-023 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235ST1 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SLD-EL-10201.DWG 02OCT25 9:04AM HOTSEAT04